



Forecasting for inventory planning: a 50-year review

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Forecasting and planning for inventory management has received considerable attention from the Operational Research (OR) community over the last 50 years because of its implications for decision making, both at the strategic level of an organization and at the operational level. Many influential contributions have been made in this area, reflecting different perspectives that have evolved in divergent strands of the literature, namely: system dynamics, control theory and forecasting theory (both statistical and judgemental). Although this pluralism is healthy in terms of knowledge advancement, it also signifies the fragmentation of the OR discipline and the lack of cross-fertilization of ideas to develop more comprehensive approaches towards the resolution of the same issues. In this paper, the relevant literature is reviewed and synthesized to promote some convergence between these different approaches to inventory forecasting and planning. The review concludes with an inter-disciplinary agenda for further research.

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Introduction

From its foundation, Operational Research (OR) has made many substantial contributions to inventory forecasting and planning. These contributions have influenced supply chain

(eg SD, forecasting, inventories research) that in turn produce different journals with, in our opinion, inadequate cross-referencing. There is great scope for their cross-utilization (Akkermans and Dellaert, 2005) to develop more compre-

